Firm Entry and AI Exposure: Evidence from US Industries

Guillermo Gallacher*

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Abstract

I study the relationship between sectoral AI exposure and firm entry in the United States. Using monthly business application data from the Census Bureau Business Formation Statistics and the AI Industry Exposure index of Felten et al. (2021), I document an association between a sector's AI exposure and its post-2022 business application growth. The association is not immediate: the effect emerges only tentatively in early 2023 before fading, then re-emerges persistently and grows monotonically from mid-2024 onwards, broadly coinciding with the diffusion and broader adoption of AI tools. While I cannot rule out confounding from other sector-specific trends, this dynamic pattern is consistent with AI progressively lowering entry costs or raising entrepreneurial returns in exposed sectors.

*Email: guillermogallacher@gmail.com. Web: guillgall.github.io.

1 Introduction

After the COVID-19 shock, aggregate business applications surged (Decker and Haltiwanger, 2023). The release of ChatGPT in November-2022 seems to have increased firm entry, too, specially after 2025, as seen in Figure 1.

This 2025 expansion is not uniform across sectors: industries such as Professional Services have increased more than the average, while others such as Construction have lagged behind. From this plot a simple observation arises: while the former sector is exposed to AI, the latter ones is less so. Does this pattern hold across all sectors, though?

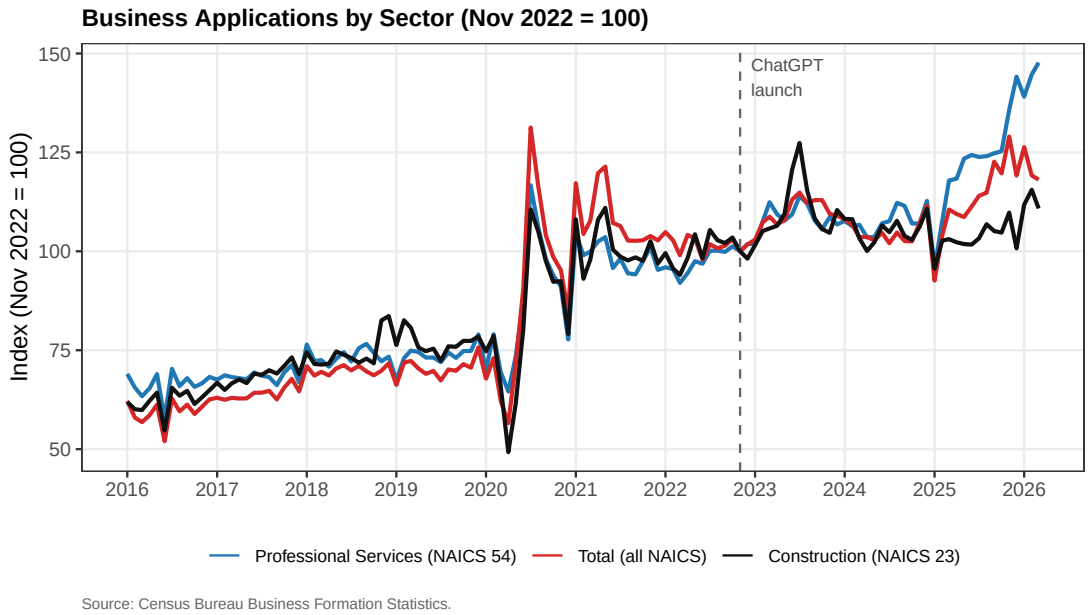

Figure 1: Business applications indexed to November 2022 = 100. Professional Services (NAICS 54), Total (all NAICS), and Construction (NAICS 23). Seasonally adjusted.

In this paper I show that it does: AI-exposed sectors have experienced a firm entry boom, while less-exposed sectors have lagged behind. The scatter-plot in Figure 2 illustrates this clearly.

I also study the timing of this relationship: I show that it held temporarily in early 2023, but then more persistently from 2024 onwards.

Contribution to the literature Most of the "economics of AI" debate focuses on workers: which jobs are at risk, and for whom. But a parallel question may be equally consequential: does AI alter the incentives to start a business?

Chatterji et al. (2025) provide a conceptual treatment of how transformative AI reshapes firm organisation, adoption incentives, and competitive dynamics—a natural complement to the reduced-form evidence presented here. Felten et al. (2021) construct the AIOE/AIIE index by combining O*NET task measures with AI capability scores. In future iterations of this paper, I plan to include other AI exposure metrics, such as the LLM-based occupational exposure scores developed by Eloundou et al. (2024).

I combine the AI literature with the firm entry dimension, by using business formation data as in Decker and Haltiwanger (2023), and papers cited therein.

2 Data

2.1 Business Formation Statistics

The Census Bureau BFS provides monthly, seasonally adjusted counts of new Employer Identification Number (EIN) applications at the 2-digit NAICS level, derived from IRS Form SS-4 filings. EIN applications are a leading indicator of actual business formation (Decker and Haltiwanger, 2023).

2.2 AI Exposure

Following Felten et al. (2021), I use the AI Industry Exposure (AIIE) index, which maps occupational AI exposure scores to 4-digit NAICS industries through BLS employment weights. I aggregate to 2-digit and 2-digit NAICS using employment-weighted averaging, with 2019 BLS OES industry employment counts as weights, consistent with Felten et al. (2021) own use of 2019 employment in constructing the index.[1] Table 6 in Appendix A reports the AIIE

[1] Results are unchanged when using an unweighted mean across 4-digit sub-industries.

2-digit score alongside summary statistics of the BFS series for each sector in the sample.

3 Results

I first focus on 2-digit NAICS industries.

3.1 Regression

Figure 2 plots the percentage change in business applications from November 2022 to the most recent month available against each sector's AI exposure.

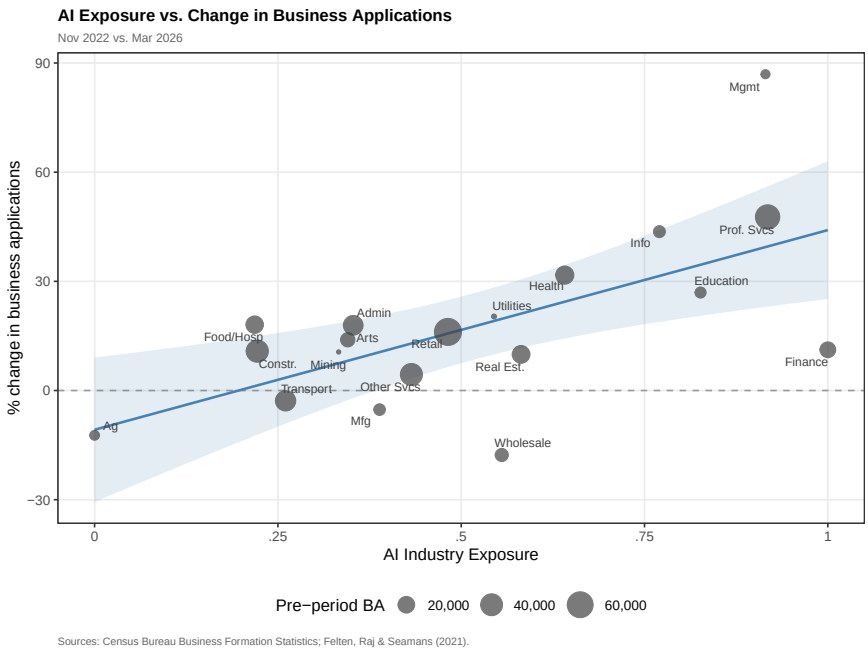

Figure 2: AI exposure versus change in business applications by 2-digit NAICS sector.

Table 1 formalises the scatter with a cross-sectional regression of the form

$$\Delta\overline{\mathrm{BA}}_i \;=\; \alpha + \delta\,\mathrm{AIIE}_i + \varepsilon_i, \tag{1}$$

where $\Delta\overline{\mathrm{BA}}_i$ is the percentage change in business applications from November 2022 to the most recent month available for sector i, and AIIE_i is the normalised AI Industry Exposure

Table 1: Cross-Sectional Regression: AI Exposure and Change in Business Applications

	OLS	WLS
AI Exposure (AIOE, normalised)	54.856***	45.673***
	(16.196)	(12.561)
Constant	-10.773	-5.947
	(9.400)	(7.085)
Sectors	19	19
R^2	0.403	0.437

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
OLS and WLS (weighted by sector mean BA, Nov 2022) estimates. Outcome: percentage change in business applications from November 2022 to the most recent month available. Each observation is a 2-digit NAICS sector ($N = 19$).

index. Column (1) is unweighted OLS; column (2) weights observations by pre-period mean business applications, giving larger sectors proportionally more influence. The slope is positive and statistically significant in both specifications, and the weighted estimate is somewhat larger, indicating that the positive relationship between AI exposure and application growth is stronger among economically larger sectors.

3.2 Difference-in-Differences

I estimate the effect of AI exposure (AIIE_i) on business applicacations (BA_{it}) using a difference-in-differences (DiD) model of the form

$$\log(\text{BA}_{it}) \;=\; \alpha_i + \gamma_t + \beta\,(\text{Post}_t \times \text{AIIE}_i) + \varepsilon_{it}, \tag{2}$$

where i indexes 2-digit NAICS sectors, t indexes calendar months, Post_t is an indicator equal to one from November 2022 onwards, α_i are sector fixed effects absorbing time-invariant differences in application rates, γ_t are month fixed effects absorbing aggregate cyclical variation, and standard errors are clustered by sector. The coefficient β identifies whether sectors with

Table 2: Difference-in-Differences Estimates

	(1) log(BA)	(2) BA Index	(3) BA Level
Post × AI Exposure	0.204	30.145	-1096.390
	(0.214)	(29.581)	(7366.702)
Observations	4959	4959	4959
R^2	0.982	0.690	0.834
Sector FE	✓	✓	✓
Month FE	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by 2-digit NAICS sector.

higher AI exposure experienced a differential change in business formation after ChatGPT.

Table 2 reports the DiD estimates. All three columns pool the entire post-November 2022 period into a single *Post* indicator, with columns (1)–(3) using log business applications, the indexed BA outcome, and raw BA levels respectively.

To translate the coefficient into economically interpretable magnitudes, Table 3 reports the implied percentage difference in business applications relative to a sector at the 25th percentile of AI exposure, evaluated at selected percentiles of the AIOE distribution. A sector at the median of AI exposure is estimated to have had 2.3% more applications than a P25 sector in the post-ChatGPT period; the gap widens to 5.1% at the 75th percentile and 10.7% at the 90th.

Table 3: Implied Effect of AI Exposure on Business Applications

Percentile	AIOE (normalised)	Effect vs. P25 sector (%)
P10	0.220	-2.4%
P25	0.337	+0.0%
P50	0.447	+2.3%
P75	0.585	+5.1%
P90	0.862	+10.7%

Notes: Implied percentage difference in business applications in the post-ChatGPT period relative to a sector at the 25th percentile of normalised AI Industry Exposure (AIOE). Computed as $(\text{AIOE}_p - \text{AIOE}_{P25}) \times \hat{\beta} \times 100$, where $\hat{\beta}$ is the DiD estimate from column (1) of Table 2.

3.3 Event Study

Figure 3 presents the event-study coefficients. The pre-period estimates ($t < 0$) are small and statistically indistinguishable from zero, supporting the parallel-trends assumption.

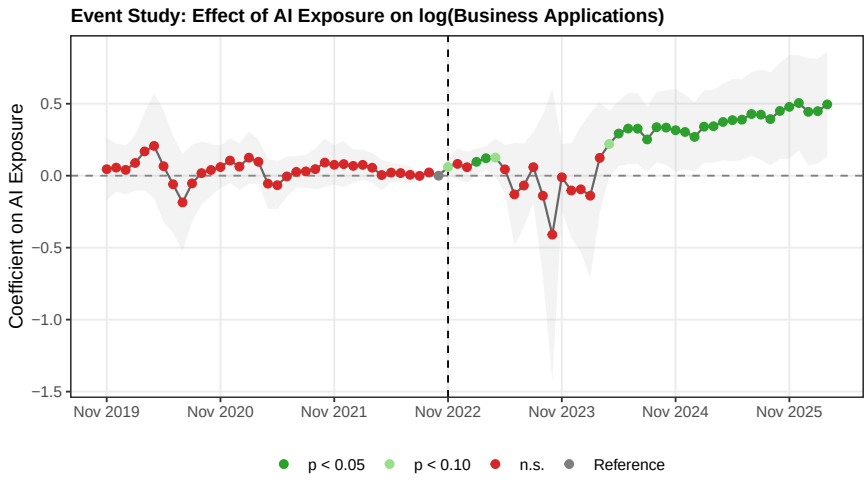

Figure 3: Event-study estimates: effect of AI exposure on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on sector-clustered standard errors.

A handful of coefficients are statistically significant in the first months immediately fol-

lowing the ChatGPT release, suggesting an early, tentative response. These effects then recede into insignificance across roughly a year, before re-emerging persistently from $t = +17$ (April 2024, $p < 0.10$) and $t = +18$ (May 2024, $p < 0.05$) onward. From May 2024, every subsequent monthly coefficient is statistically significant at the 5 % level or better, and point estimates grow monotonically through the end of the sample, reaching 0.48 at $t = +36$.

4 Robustness Checks

4.1 Alternative Business Applications Metrics

I probe robustness by substituting the outcome with the full suite of available BFS series spanning the formation funnel.

On the applications side, I use corrected business applications (CBA), which purge filings unlikely to yield an employer business; high-propensity applications (HBA), which restrict to applicants with a high predicted probability of hiring; and applications with planned wages (WBA), indicating explicit intent to hire.

On the formations side, I use seasonally-adjusted formations within four and eight quarters (SBF4Q, SBF8Q). The realised formation counts (BF4Q, BF8Q) are excluded as the Census has not yet published these beyond 2022 due to publication lags.

Figure 4 presents the corresponding event-study coefficients for all six series. The pre-period estimates are small and insignificant across every panel, ruling out differential pre-trends. Post-launch dynamics are qualitatively consistent: coefficients drift upward at similar timing and reach statistical significance in the same window regardless of which outcome is used.

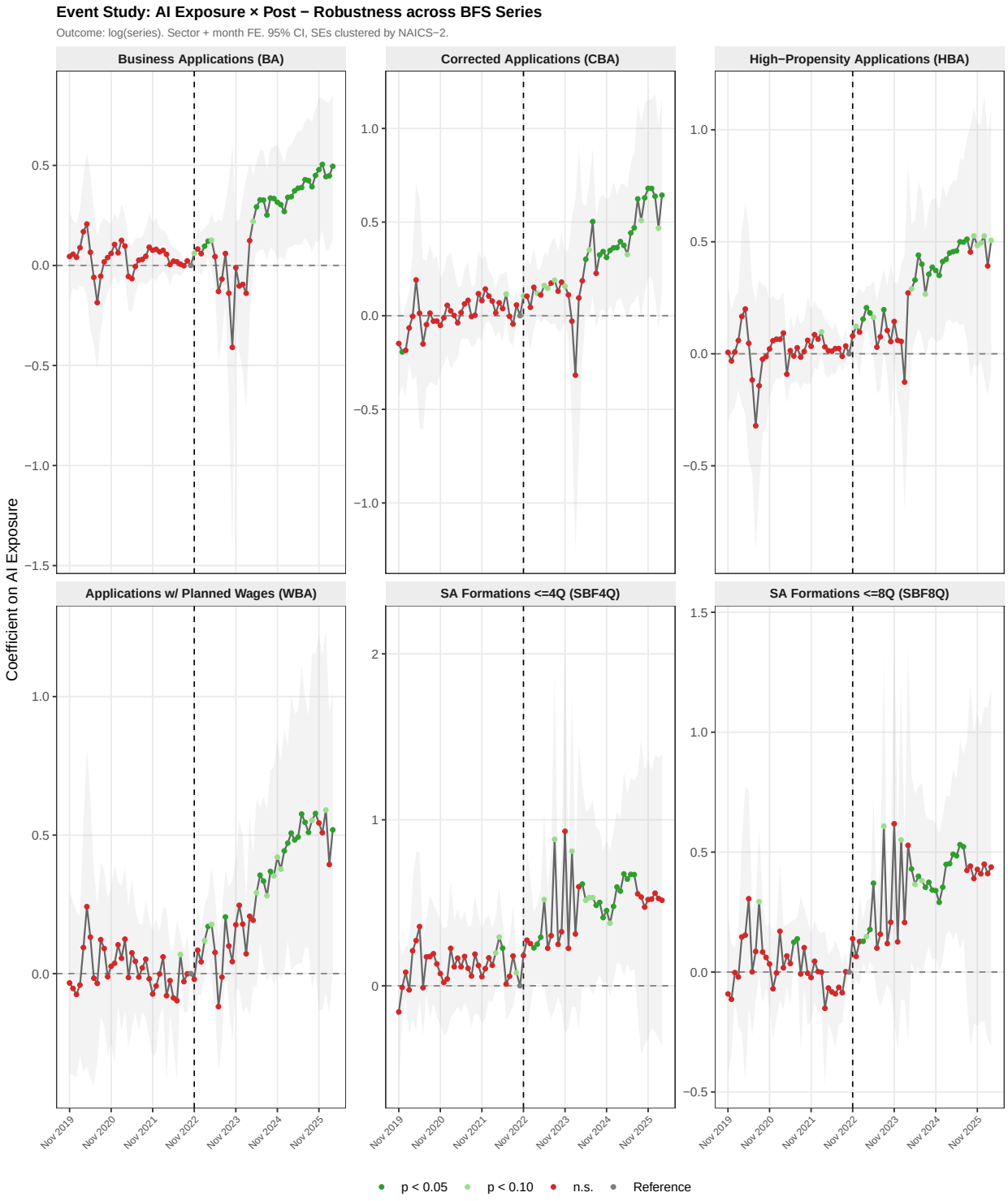

Figure 4: Event-study estimates for six BFS series. Dashed vertical line marks November 2022 ($t = 0$).

Table 4: Robustness: Alternative BFS Series

	(1) log(BA)	(2) log(CBA)	(3) log(HBA)	(4) log(WBA)	(5) log(SBF4Q)	(6) log(SBF8Q)
Post × AI Exposure	0.204	0.438	0.247	0.269	0.594	0.444
	(0.214)	(0.287)	(0.283)	(0.263)	(0.352)	(0.287)
Observations	4959	4959	4959	4959	4951	4744
R^2	0.982	0.980	0.983	0.984	0.983	0.983
Sector FE	✓	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01

Standard errors clustered by 2-digit NAICS sector. All specifications include sector and month fixed effects. BA = total business applications; CBA = corrected business applications; HBA = high-propensity applications; WBA = applications with planned wages; SBF4Q/SBF8Q = seasonally-adjusted formations within 4/8 quarters.

4.2 3-digit NAICS

Figure 5 replicates the motivating scatter at the finer 3-digit NAICS level, comparing mean weekly business applications in November 2022 with November 2025 across 75 sectors. The positive cross-sectional gradient visible at the 2-digit level (Figure 2) persists at higher disaggregation, and the bubble sizes — proportional to pre-period application volume — make clear that larger sectors tend to sit higher on the fitted line. Table **??** formalises this. The unweighted OLS slope is near zero and statistically insignificant ($\hat{\delta} = 0.9$, $p > 0.10$, $R^2 \approx 0.00$), reflecting substantial noise among the many small 3-digit sectors. Once sectors are weighted by their pre-period application volume, however, the slope rises sharply to 41.1 ($p < 0.01$, $R^2 = 0.31$): among the sectors that account for the bulk of economic activity, higher AI exposure is strongly associated with faster application growth.

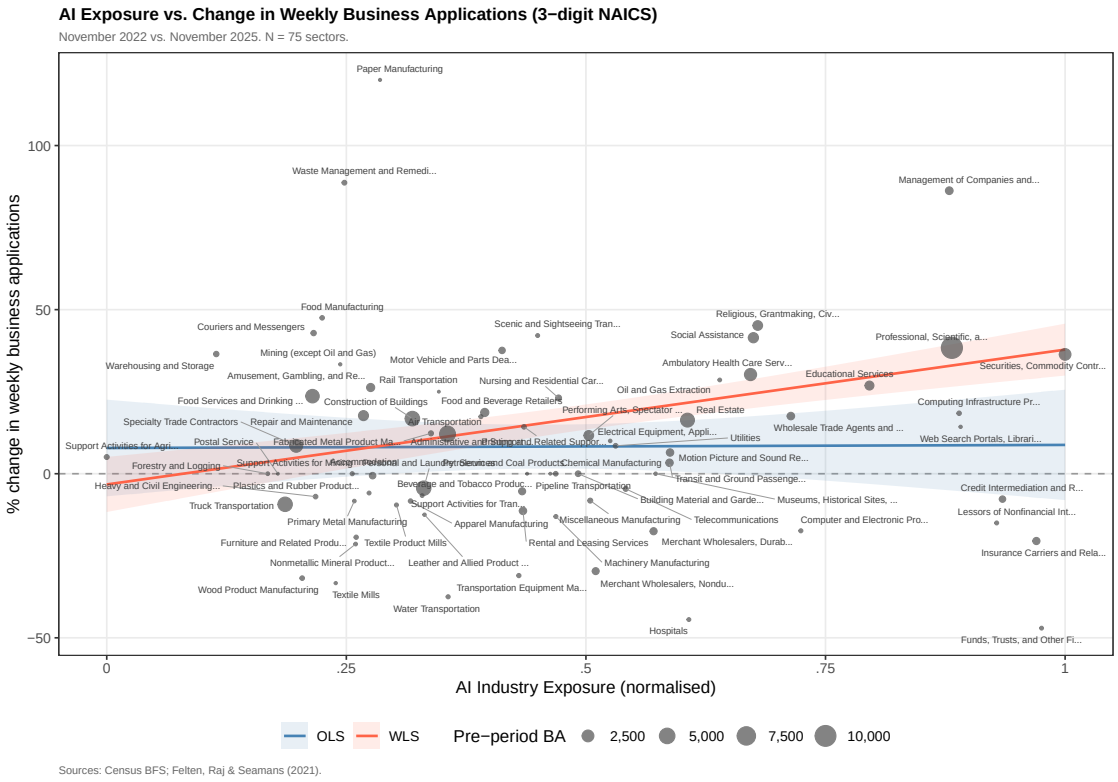

Figure 5: AI exposure versus change in weekly business applications, 3-digit NAICS sectors. November 2022 vs. November 2025. Bubble size proportional to pre-period mean weekly applications. Blue line: unweighted OLS. Red line: WLS weighted by pre-period application volume. Shaded bands: 95 % confidence intervals.

Table 5: Cross-Sectional Regression: AI Exposure vs. Change in Weekly Applications (3-digit NAICS)

	OLS	WLS
AI Exposure (AIOE, normalised)	0.936	41.092***
	(14.332)	(7.223)
Constant	7.840	-3.271
	(7.394)	(4.212)
Sectors (3-digit)	75	75
R^2	0.000	0.307

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
OLS and WLS (weighted by pre-period mean weekly BA) at 3-digit NAICS. Outcome: percentage change in mean weekly business applications from November 2022 to November 2025.

Figure 6 presents the event-study coefficients from estimating equation (2) on the weekly 3-digit NAICS panel. Because the raw weekly series exhibit strong sector-specific seasonality — high-AIOE sectors such as Professional Services have a markedly different within-year cycle than low-AIOE sectors such as Construction — each sector's weekly BA series is first seasonally adjusted using STL decomposition on the log scale (multiplicative, periodic window) before entering the regression. The event study uses 4-week bins, sector and ISO year-week fixed effects, and observations weighted by each sector's pre-period mean weekly BA. Weighting by sector size is motivated by the cross-sectional evidence in Table **??**: the relationship between AI exposure and application growth is concentrated among larger sectors, so weighting ensures that identification comes from the same margin. The pre-period coefficients are small and statistically indistinguishable from zero, supporting the parallel-trends assumption at finer sectoral disaggregation. Post-launch dynamics mirror those in the 2-digit baseline: coefficients drift upward and eventually reach statistical significance, consistent with AI-exposed sectors experiencing a relative surge in weekly business applications following the ChatGPT release.

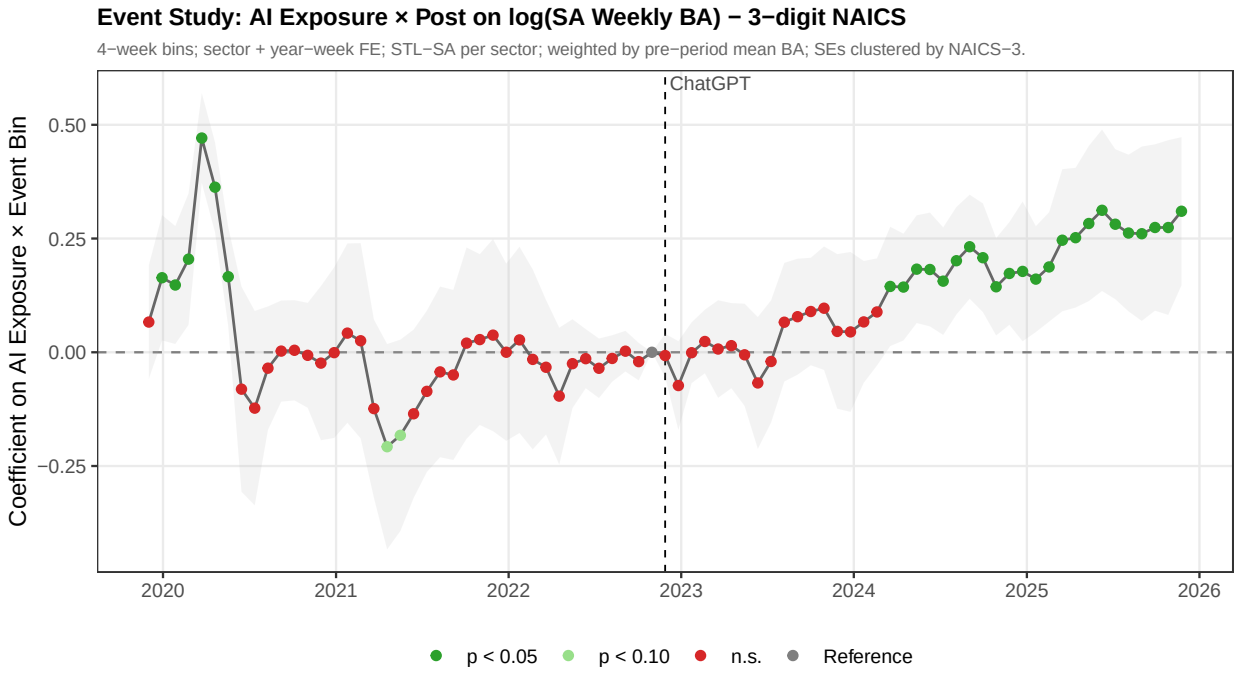

Figure 6: Event-study estimates at 3-digit NAICS. Outcome: log seasonally adjusted weekly business applications (STL decomposition, multiplicative, per sector). 4-week event-time blocks relative to ChatGPT launch (28 Nov 2022). Reference block: $t \in [-4, -1]$. Observations weighted by pre-period mean weekly BA. Sector and ISO year-week fixed effects; SEs clustered by NAICS-3.

4.3 US States

Appendix C reports a state-level robustness check using the AI Geographic Exposure (AIGE) index of Felten et al. (2021) and state × month panels. The state-level estimates are statistically insignificant, likely because state aggregation dilutes the within-sector AI variation that drives the main results; the sector-level estimates remain the primary evidence.

5 Conclusion

[Conclusion placeholder.]

References

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A Summary Statistics

Table 6: AI Industry Exposure and Business Formation Statistics by Sector

Sector	NAICS	AIIE	Mean	SD	Min	Max	N
			Monthly BA (thousands)				(months)
Finance & Insurance	52	2.04	14.0	3.6	8.4	22.0	261
Professional, Scientific, Technical Svcs	54	1.71	38.2	11.7	19.8	76.7	261
Management of Companies	55	1.70	2.2	1.3	0.9	6.6	261
Educational Services	61	1.34	4.6	1.5	2.4	8.6	261
Information	51	1.11	6.1	1.8	3.0	11.4	261
Health Care & Social Assistance	62	0.59	18.3	5.6	10.2	33.1	261
Real Estate & Rental	53	0.35	16.9	5.4	8.9	26.5	261
Wholesale Trade	42	0.24	8.9	1.3	6.1	16.2	261
Utilities	22	0.20	0.3	0.2	0.1	0.8	261
Retail Trade (NAICS 44–45)	RET	-0.05	45.2	24.7	20.1	131.0	261
Other Services	81	-0.25	26.6	10.0	12.7	62.8	261
Manufacturing (NAICS 31–33)	MNF	-0.43	5.7	1.3	3.7	10.5	261
Admin. & Support Services	56	-0.58	20.3	7.2	10.9	41.9	261
Arts, Entertainment, & Recreation	71	-0.61	9.5	2.2	5.3	15.4	261
Mining, Quarrying, Oil & Gas	21	-0.66	0.6	0.1	0.3	1.4	261
Transportation & Warehousing (48–49)	TW	-0.95	18.8	9.7	7.9	44.0	261
Construction	23	-1.10	29.7	9.1	16.0	52.9	261
Accommodation & Food Services	72	-1.12	15.6	5.5	8.5	31.6	261
Agriculture, Forestry, Fishing	11	-2.00	3.5	1.5	1.9	19.4	261

Notes: AIIE = AI Industry Exposure index from Felten et al. (2021), constructed from O*NET task descriptions and aggregated from 4-digit to 2-digit NAICS by employment-weighted averaging. Mean, SD, Min, and Max refer to monthly seasonally adjusted business applications (thousands) over the full sample period (January 2006 to early 2026). N is the number of monthly observations. Sectors sorted by AIIE score descending.

B Event Study via Cross-Sectional Regressions

As a robustness check on the DiD event study in Section 3.3, I re-estimate the dynamic effect of AI exposure on business applications without the two-way fixed-effects structure, running instead a separate cross-sectional regression for each calendar month:

$$\Delta \log \mathrm{BA}_{i,t} \;=\; \alpha_t + \beta_t \,\mathrm{AIIE}_i + \varepsilon_{i,t}, \tag{3}$$

where $\Delta \log \mathrm{BA}_{i,t} = \log(\mathrm{BA}_{i,t}) - \log(\mathrm{BA}_{i,\text{Nov 2022}})$ is the log-change in business applications for sector i between month t and the November 2022 baseline, and AIIE_i is the normalised AI Industry Exposure index. Each of the 19 two-digit NAICS sectors contributes one observation per regression; November 2022 is the omitted reference with $\widehat{\beta}_t = 0$ by construction. Standard errors are HC1 heteroskedasticity-robust.

The coefficient path $\{\widehat{\beta}_t\}$, plotted in Figure 7, is the analogue of a standard event-study but estimated without pooling across months and without imposing sector or month fixed effects: identification comes purely from the cross-sectional gradient between AI exposure and application growth at each horizon. The pre-period estimates cluster tightly around zero, and the post-launch dynamics closely track those of the DiD event study in Figure 3: a short-lived response in early 2023, a flat period through late 2023, and a persistently significant, monotonically widening gap from mid-2024 through the end of the sample in March 2026.

Table 7: Cross-Sectional Event Study (Spec. 1: OLS, outcome Δ log

	May 2023 (t=+6)	Nov 2023 (t=+12)	May 2024 (t=+18)	Nov 2024 (t=+24)
AI Exposure (AIOE)	-0.017	-0.072	0.232**	0.255**
	(0.058)	(0.097)	(0.083)	(0.111)
Constant	0.069*	0.128*	-0.099**	-0.060
	(0.036)	(0.071)	(0.043)	(0.052)
Sectors	19	19	19	19
R^2	0.007	0.022	0.246	0.249

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Each column is a separate cross-sectional OLS regression across 2-digit NAICS sectors in the month ind
November 2022 baseline (the reference month). HC1 heteroskedasticity-robust standard errors in parent

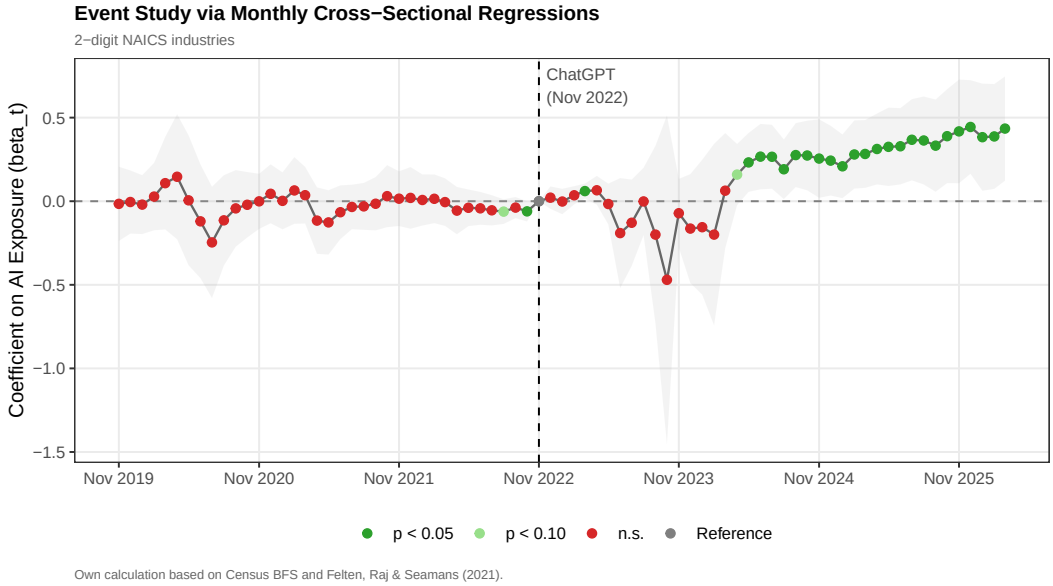

Figure 7: Monthly cross-sectional regressions of the log-change in business applications (relative to November 2022) on AI Industry Exposure across 2-digit NAICS sectors. Each point is the OLS slope $\widehat{\beta}_t$ from equation (3) estimated in a separate regression for that month; shaded band is the 95 % confidence interval using HC1 robust standard errors. Points coloured by significance.

Table 7 reports the underlying regressions at selected post-launch horizons (+6, +12, +18, +24, +30, +36 months).

Figure 8 repeats the exercise under three additional specifications: weighted least squares using pre-period business applications as sector weights; an OLS specification that replaces

the continuous AIIE score with AI-exposure quartile dummies (Q4 vs. Q1 reference); and an OLS specification that uses %-change in BA as the outcome rather than log-change. The coefficient paths are qualitatively similar across all four specifications, with post-2024 effects that are statistically significant and widening in every case.

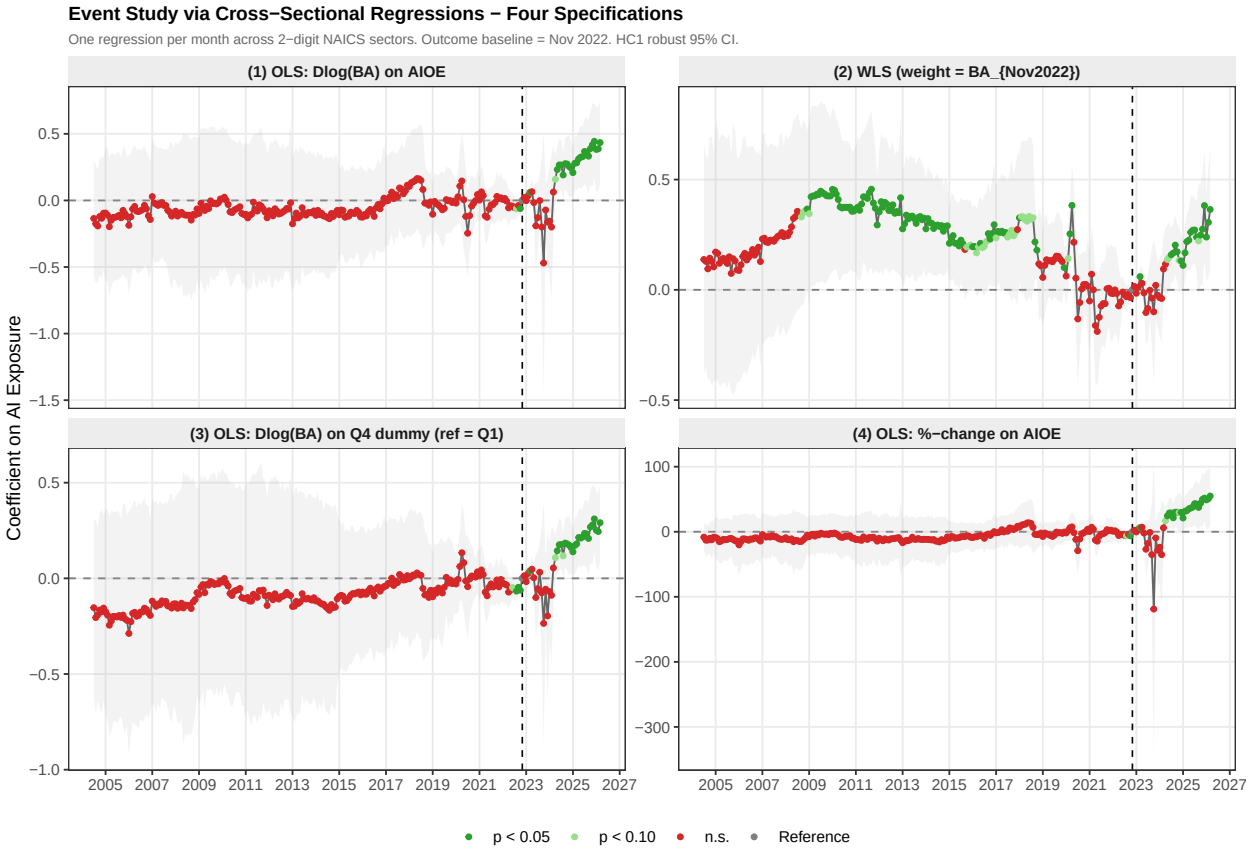

Figure 8: Robustness of the cross-sectional event study across four specifications. Panel (1) is the baseline OLS on normalised AIIE; panel (2) reweights by pre-period BA; panel (3) uses AI-exposure quartile dummies; panel (4) replaces the log-change outcome with a %-change outcome. Shaded bands are 95 % confidence intervals using HC1 robust standard errors.

C State-Level Robustness

Table 8 presents the state-level DiD estimates. The unit of observation is a state-month cell, with the outcome being log total seasonally adjusted business applications aggregated across all sectors within the state. The AI exposure measure is the AIGE index of Felten

et al. (2021), which captures the AI exposure of a state's workforce through its occupational mix. State fixed effects absorb permanent cross-state differences in entrepreneurial culture, regulatory environment, and industrial composition; month fixed effects absorb national macro shocks as before. Standard errors are clustered at the state level (56 clusters).

Figure 9 provides raw descriptive evidence at the state level, analogous to Figure **??** for sectors. States are sorted into quartiles by their AIGE index and the cross-state mean of indexed business applications (January 2020 = 100) is plotted for each quartile.

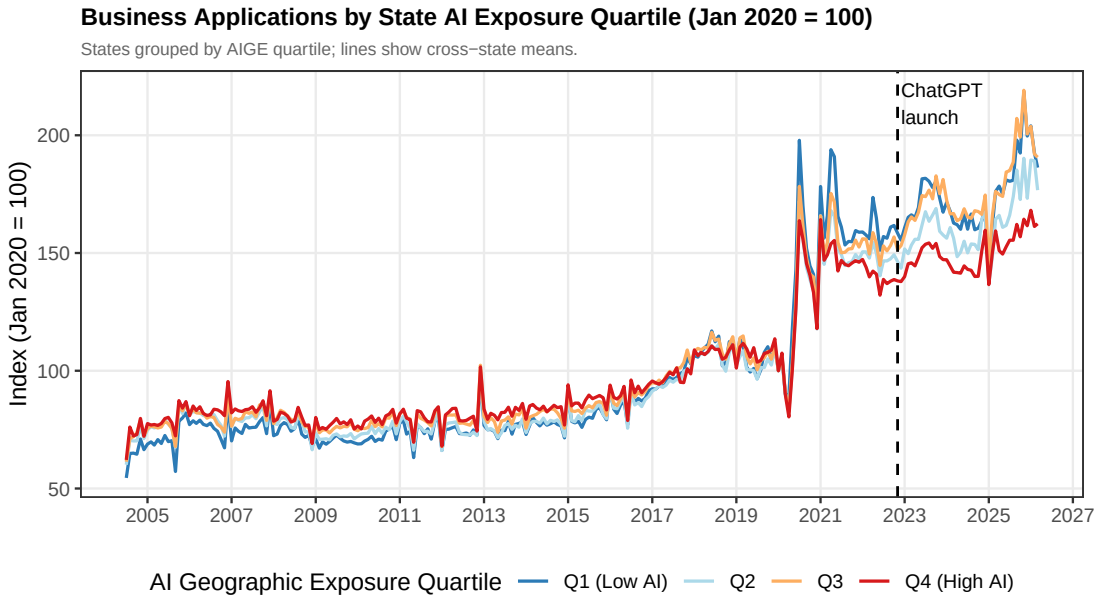

Figure 9: Mean business applications (index, January 2020 = 100) by AI Geographic Exposure (AIGE) quartile across US states and territories. Q1 denotes the lowest-exposure quartile; Q4 the highest. Lines show simple cross-state means within each quartile. Dashed vertical line marks November 2022 (ChatGPT launch).

Column (1) shows the single-period estimate; column (2) uses the indexed BA outcome; column (3) splits the post period as in the main analysis. The state-level estimates are statistically insignificant across all specifications, in contrast to the sector-level results in Table 2.

Figure 10 shows the corresponding event-study coefficients, and Figure 11 repeats the exercise across the four BFS series published at the state level (BA, HBA, WBA, CBA;

Table 8: State-Level Difference-in-Differences Estimates

	(1) log(BA)	(2) BA Index	(3) Two-period
$\text{Post}_{\text{ChatGPT}} \times$ AI Exposure	-0.445	-67.191	-0.423
	(0.274)	(42.331)	(0.265)
Observations	13572	13572	13572
R^2	0.986	0.825	0.986
State FE	✓	✓	✓
Month FE	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by state. AI exposure is the AI Geographic Exposure (AIGE) index from Felten et al. (2021) Appendix C, constructed from occupation-level AIOE scores weighted by the state employment-mix across occupations. $\text{Post}_{\text{ChatGPT}}$: Nov 2022 – Apr 2025. $\text{Post}_{\text{Agentic}}$: May 2025 onwards.

formation series are national-only). Pre-period coefficients are flat across all panels, consistent with parallel pre-trends, but the post-launch coefficients are imprecise and do not reach conventional significance levels in any series.

The absence of a detectable state-level effect likely reflects two compounding limitations. First, states are highly diversified: even a state with a high-AIGE workforce will contain many low-AI sectors that dilute the aggregate application signal. The sector-level design exploits within-sector variation in AI exposure that is averaged away when outcomes are aggregated to the state level. Second, the cross-state dispersion of AIGE is considerably smaller than the cross-sector dispersion of AIOE, reducing statistical power. Together, these features make the state-level panel a demanding test; the sector-level estimates remain the primary evidence.

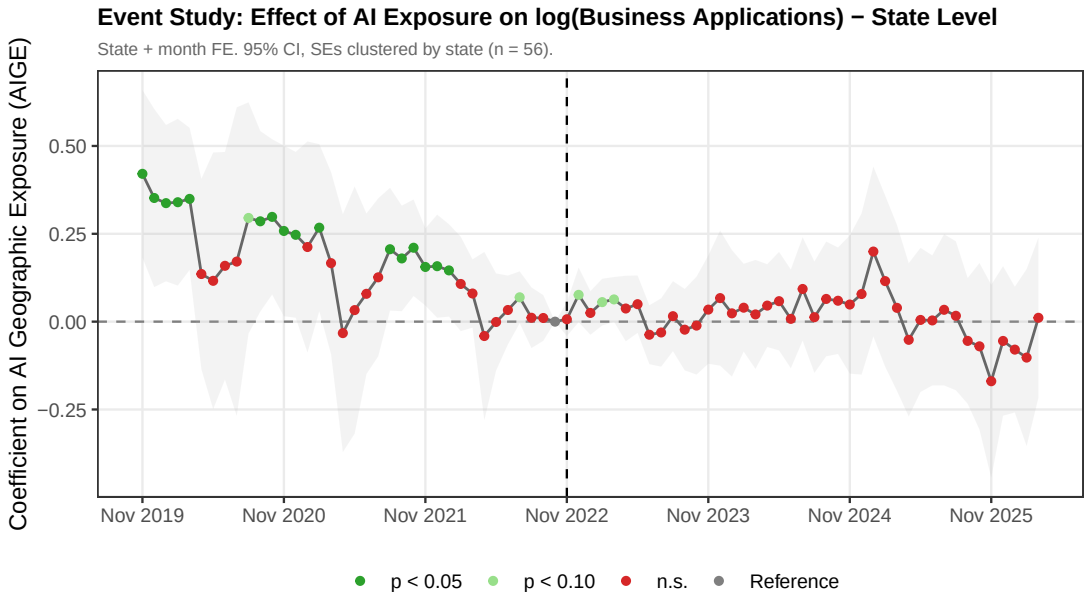

Figure 10: State-level event-study estimates: effect of AI Geographic Exposure (AIGE) on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on state-clustered standard errors (56 clusters). State and month fixed effects included. Coefficients are statistically insignificant throughout.

Figure 11 replicates the state-level event study across the four BFS series available at state level.

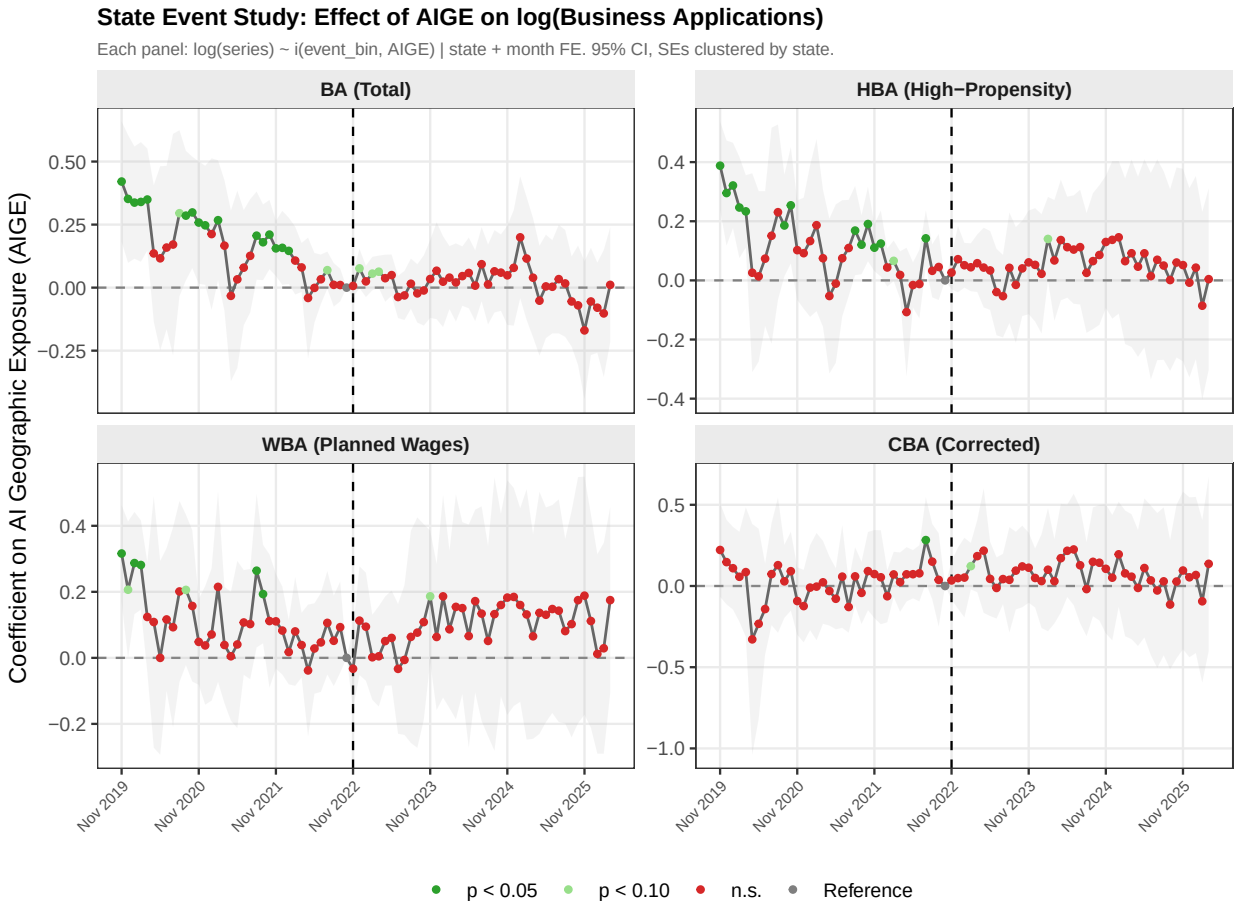

Figure 11: State-level event-study estimates for four BFS series. Each panel plots the interaction coefficient between AI Geographic Exposure (AIGE) and a month-relative-to-November-2022 indicator, estimated on the state × month panel. BA = total applications; HBA = high-propensity applications; WBA = applications with planned wages; CBA = corrected applications. Business formation series (BF4Q, BF8Q, etc.) are not published at state level by the Census Bureau. Shaded bands are 95 % confidence intervals based on state-clustered standard errors (56 clusters). All specifications include state and month fixed effects. Dashed vertical line marks November 2022 ($t = 0$).